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Herbert RD, de Noronha M, Kamper SJ

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Stretching to prevent or reduce muscle soreness after exercise (Review)

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Stretching to prevent or reduce muscle soreness after exercise

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ABSTRACT

Background

Many people stretch before or after engaging in athletic activity. Usually the purpose is to reduce risk of injury, reduce soreness after exercise, or enhance athletic performance. This is an update of a Cochrane review first published in 2007.

Objectives

The aim of this review was to determine effects of stretching before or after exercise on the development of delayed-onset muscle soreness.

Search methods

We searched the Cochrane Bone, Joint and Muscle Trauma Group Specialised Register (to 10 August 2009), the Cochrane Central Register of Controlled Trials (2010, Issue 1), MEDLINE (1966 to 8th February 2010), EMBASE (1988 to 8th February 2010), CINAHL (1982 to 23rd February 2010), SPORTDiscus (1949 to 8th February 2010), PEDro (to 15th February 2010) and reference lists of articles.

Selection criteria

Eligible studies were randomised or quasi-randomised studies of any pre-exercise or post-exercise stretching technique designed to prevent or treat delayed-onset muscle soreness (DOMS). For the studies to be included, the stretching had to be conducted soon before or soon after exercise and muscle soreness had to be assessed.

Data collection and analysis

Risk of bias was assessed using The Cochrane Collaboration's 'Risk of bias' tool and quality of evidence was assessed using GRADE. Estimates of effects of stretching were converted to a common 100-point scale. Outcomes were pooled in fixed-effect meta-analyses.

Main results

Twelve studies were included in the review. This update incorporated two new studies. One of the new trials was a large field-based trial that included 2377 participants, 1220 of whom were allocated stretching. All other 11 studies were small, with between 10 and 30 participants receiving the stretch condition. Ten studies were laboratory-based and other two were field-based. All studies were exposed to either a moderate or high risk of bias. The quality of evidence was low to moderate.

There was a high degree of consistency of results across studies. The pooled estimate showed that pre-exercise stretching reduced soreness at one day after exercise by, on average, half a point on a 100-point scale (mean difference -0.52, 95% CI -11.30 to 10.26; 3 studies). Post-exercise stretching reduced soreness at one day after exercise by, on average, one point on a 100-point scale (mean difference -1.04, 95% CI -6.88 to 4.79; 4 studies). Similar effects were evident between half a day and three days after exercise. One large study showed that stretching before and after exercise reduced peak soreness over a one week period by, on average, four points on a 100-point scale (mean difference -3.80, 95% CI -5.17 to -2.43). This effect, though statistically significant, is very small.

Authors' conclusions

The evidence from randomised studies suggests that muscle stretching, whether conducted before, after, or before and after exercise, does not produce clinically important reductions in delayed-onset muscle soreness in healthy adults.

PLAIN LANGUAGE SUMMARY

Stretching to prevent or reduce muscle soreness after exercise

Many people stretch prior to or after engaging in physical activities such as sport. Usually the purpose is to reduce the risk of injury, reduce soreness after exercise, or enhance athletic performance. This review looked at the effects of stretching on muscle soreness only.

The review located 12 relevant randomised controlled studies looking at the effect of stretching before or after physical activity on muscle soreness. Eleven studies were small with between 10 to 30 people being allocated stretching exercises. In contrast, one study was large with 2337 participants, 1220 of whom were in the stretching group. Ten studies were conducted in laboratories using standardised exercises. The only two studies, which included the only large study, were so-called field-based studies. These examined the effect of stretching on muscle soreness associated with self-selected physical activity. The studies were of low to moderate quality. Some of the studies examined the effects of stretching before physical activity, some examined the effects of stretching after physical activity, and some examined effects of stretching both before and after physical activity.

The studies produced very consistent findings. They showed there was little or no effect of stretching on the muscle soreness experienced in the week after the physical activity.

SUMMARY OF FINDINGS FOR THE MAIN COMPARISON *[Explanation]*

Pre-exercise stretching to prevent soreness						
Patient or population: people doing unaccustomed exercise Settings: any Intervention: pre-exercise stretching						
Outcomes	Illustrative comparative risks* (95% CI)		Relative effect (95% CI)	No of Participants (studies)	Quality of the evidence (GRADE)	Comments
	Assumed risk	Corresponding risk				
	Control	pre-exercise stretching				
Soreness on day 1 Scale from: 0 to 100. Follow-up: 1 day	The mean soreness on day 1 in the control groups was 49	The mean soreness on day 1 in the intervention groups was 0.52 points lower (11.30 lower to 10.26 higher)		70 (3 studies)	⊕⊕○○ low ^{1,2,3}	
Soreness on day 2 Scale from: 0 to 100. Follow-up: 2 days	The mean soreness on day 2 in the control groups was 51	The mean soreness on day 2 in the intervention groups was 0.72 points higher (11.20 lower to 12.64 higher)		40 (2 studies)	⊕⊕○○ low ^{1,2,3}	
*The assumed risk for the 'control' group is based on the mean value of the results for all control groups in the trials included in each analysis. The corresponding risk (and its 95% confidence interval) is based on the assumed risk in the comparison group and the relative effect of the intervention (and its 95% CI). CI: Confidence interval;						

GRADE Working Group grades of evidence

High quality: Further research is very unlikely to change our confidence in the estimate of effect.

Moderate quality: Further research is likely to have an important impact on our confidence in the estimate of effect and may change the estimate.

Low quality: Further research is very likely to have an important impact on our confidence in the estimate of effect and is likely to change the estimate.

Very low quality: We are very uncertain about the estimate.

- ¹ Studies were randomised but little or no information was provided about allocation concealment, loss to follow-up or analysis by intention to treat.
- ² Most or all of these studies involved inducing muscle soreness using a standardised protocol. Such studies provide an indirect evaluation of the effects of stretching on soreness caused by recreational exercise.
- ³ There were too few studies to reliably assess risk of publication bias or other sources of small study bias.

BACKGROUND

Many people stretch before or after (or before and after) engaging in athletic activity. Usually the purpose is to reduce the risk of injury, reduce soreness after exercise, or enhance athletic performance (Cross 1999; de Vries 1961; Gleim 1990; Gleim 1997). This review focuses on the effects of muscle stretching on muscle soreness. Another Cochrane review (Yeung 2003) has examined whether stretching prevents soft-tissue injury, and several non-Cochrane systematic reviews have examined whether stretching prevents muscle soreness or injury, or enhances performance (Andersen 2005; Fradkin 2006; Herbert 2002; Rubini 2007; Shrier 2004; Small 2008; Thacker 2004; Weldon 2003).

The muscle soreness that is the focus of this review is sometimes called delayed-onset muscle soreness to differentiate it from soreness that occurs as muscle fatigue or immediately after muscles have been fatigued. Delayed-onset muscle soreness (hereafter referred to simply as muscle soreness) is usually caused by unaccustomed exercise, particularly exercise that requires primarily eccentric muscle action. Eccentric action occurs when a muscle generates tension to control the rate it lengthens. Typically the soreness arises within the first day after exercise and peaks in intensity at around 48 hours (Balnave 1993; Bobbert 1986; Wessel 1994). The soreness is usually associated with tenderness and may be associated with swelling (Bobbert 1986; Wessel 1994).

The series of events that ultimately cause muscle soreness are only partly understood. The initial event is probably mechanical disruption of sarcomeres (contractile units within muscle fibres; Proske 2001). This causes swelling of damaged muscle fibres and initiates an inflammatory response, which could excite nociceptors (receptors capable of transmitting information about pain; Lieber 2002). Muscle soreness is usually only experienced when the muscle contracts or is stretched. This indicates that, while swelling of muscle fibres and inflammation may be necessary for muscle soreness to occur, swelling and inflammation are not sufficient to cause soreness on their own. Muscle soreness is exacerbated by vibration over the belly of the muscle and pressure thresholds increase when large diameter afferents are blocked (large diameter afferents are nerve cells that conduct information about touch and limb position). This indicates that excitation of large diameter muscle afferents, probably primary spindle afferents from stretch receptors in muscles, is involved in generating the sensation of soreness (Weerakkody 2001).

The practice of stretching to prevent muscle soreness was encouraged by early investigators of muscle soreness who thought that unaccustomed exercise caused muscle spasm (de Vries 1961). Muscle spasm was believed to impede blood flow to the muscle, causing ischaemic pain and further spasm. Stretching the muscle was thought to restore blood flow to the muscle and interrupt the pain-spasm-pain cycle. The muscle spasm theory of muscle soreness has since been discredited (Bobbert 1986) but the practice of stretching persists.

Typically people who stretch to prevent muscle soreness do so prior to exercise, but some people stretch after exercise. Usually each at-risk muscle is stretched for between 15 seconds and two minutes, once or several times. Some proponents of stretching recommend applying a sustained stretch to the relaxed muscle (de Vries 1961; this is called static stretching), but others recommend more elaborate techniques such as the 'contract-relax-agonist contract' technique (Feland 2001). This technique and related techniques (collectively called proprioceptive neuromuscular facilitation (PNF) techniques) involve contracting the muscle strongly before applying a stretch.

Why it is important to do this review

This is an update of a Cochrane review first published in 2007 (Herbert 2007). The earlier review concluded that the evidence derived from small, mainly laboratory-based studies of stretching indicated that pre-exercise or post-exercise muscle stretching does not reduce delayed-onset muscle soreness in young healthy adults. A large randomised trial (2,377 participants) of both pre-exercise and post-exercise stretching has been published since then.

OBJECTIVES

This review aims to determine the effect of stretching before or after exercise on the development of post-exercise muscle soreness.

METHODS

Criteria for considering studies for this review

Types of studies

We considered any randomised or quasi-randomised (method of allocating participants to a treatment that is not strictly random; e.g. by date of birth, hospital record number and alternation) studies of pre-exercise or post-exercise stretching techniques designed to prevent delayed-onset muscle soreness.

Types of participants

Trials involving participants of any age group and of either sex.

Types of interventions

Any pre-exercise or post-exercise stretching technique designed to prevent delayed-onset muscle soreness, provided the stretching was conducted soon before or soon after exercise of any type.

Types of outcome measures

Measures of muscle soreness (pain).

Search methods for identification of studies

Electronic searches

We searched the Cochrane Bone, Joint and Muscle Trauma Group Specialised Register (10 August 2009), the Cochrane Central Register of Controlled Trials (OVID WEB; 2010, Issue 1; [Appendix 1](#)), MEDLINE (OVID WEB; 1966 to 8 February 2010; [Appendix 1](#)), EMBASE (1988 to 8 February 2010; [Appendix 1](#)), CINAHL (EBSCO; 1982 to 23 February 2010; [Appendix 1](#)), SPORTDiscus (EBSCO; 1949 to 8 February 2010; [Appendix 1](#)), and the Physiotherapy Evidence Database PEDro (accessed 15 February 2010; [Appendix 1](#)). We also searched the search the [WHO International Clinical Trials Registry Platform](#) for ongoing and recently completed trials (October 2010). There were no restrictions based on publication status or language.

Searching other resources

We searched reference lists of articles.

Data collection and analysis

One of the authors (RH) of this review is an author of one of the included studies ([Jamtvedt 2010](#)). Independent data extraction and assessment of risk of bias for this trial were carried out by the other two review authors.

Selection of studies

Two authors screened search results for potentially eligible studies. When titles and abstracts suggested a study was potentially eligible for inclusion, a paper copy of the report was obtained. Disagreements between the two authors regarding a study's eligibility were resolved by discussion or, where necessary, by the third author.

Data extraction and management

Soreness data from stretch and comparison groups were extracted from study reports using a pro forma. This was done by the authors; again, disagreements were resolved by discussion or by the third author. Some studies reported raw scores, whereas others report change scores or covariance-adjusted scores. It was also expected that outcomes would usually be measured at a series of times following exercise. Where available, data were extracted at one-day daily intervals following exercise. Only one outcome was recorded from each study for each time. Where data were available in more than one format, covariate-adjusted data were most preferred, then change scores, then raw scores.

Assessment of risk of bias in included studies

Risk of bias was assessed using The Cochrane Collaboration's 'Risk of bias' tool. Quality of evidence was assessed using GRADE. Two authors independently assessed each study. Disagreements were resolved by discussion. Where necessary, the third author was consulted. It was anticipated that several of the included studies would be laboratory-based studies which were unlikely to describe elements thought to be important in conventional clinical trials. These studies may appear methodologically weak when assessed with a tool designed for clinical trials. Nonetheless it was thought useful to document the degree to which they conform with best practice clinical trial design.

Unit of analysis issues

Where within-participant studies (studies in which participants' limbs were allocated to stretch and control conditions) or cross-over studies (studies in which each participant was sequentially exposed to both stretch and control conditions) did not provide estimates of the variance of differences between conditions or of the correlations between outcomes under the two conditions being compared ([Elbourne 2002](#)), the variance of the difference between conditions was estimated by assuming the data were independent.

Data synthesis

Pain data were typically reported on 100 mm visual analogue scales or 10-point scales. To facilitate pooling data were converted to a common 100-point scale. The effect of stretching was estimated from each study using the difference between mean pain levels in the stretch and comparison groups. Where there was apparent qualitative homogeneity in participants, interventions and outcome measures, meta-analysis was considered. The effects of stretching were described as mean differences. Data were inspected for statistical heterogeneity. The fixed-effect model was used to pool findings across studies where there was no evidence of heterogeneity of effect (Cochran Q, $P < 0.1$ and low I^2 ; [Fleiss 1993](#); [Higgins 2002](#)). Effects of pre-exercise and post-exercise stretching were initially estimated separately but, as there was no evidence of heterogeneity, they were combined in a secondary analysis. A post-hoc analysis was conducted to obtain separate estimates from laboratory-based studies and the two community-based studies. An additional post-hoc analysis was conducted to obtain separate estimates from studies of static stretch and the one study that used another method of stretching.

RESULTS

Description of studies

The search retrieved 43 potentially eligible studies of which 12 were eligible for inclusion in the review (Buroker 1989; Dawson 2005; Gulick 1996; High 1989; Jamtvedt 2010; Johansson 1999; Marques 2006; Maxwell 1988; McGlynn 1979; Terry 1985; Wessel 1994a; Wessel 1994b). The titles of a further four papers, all of which had initially been located with SPORTDiscus, appeared potentially eligible but have not yet been retrieved (Buckenmeyer 1998; Costa 1997; Lin 1999; Lu 1992), and thus still await assessment.

Included studies

Two new studies (one small and one large) were added in this update of the review. Most studies were small (less than 30 participants in the intervention group) but one study (Jamtvedt 2010) involved 2377 participants, 1220 of whom were in the intervention group. In all but one study participants were young healthy adults (age less than 40 years or mean age less than 26 years); only Jamtvedt 2010 had substantial representation of older adults (mean age 39.9 years, SD 12.6). All but two of the studies were carried out in laboratory settings and involved standardised exercise. The exceptions investigated the effect of post-game stretching on muscle soreness in Australian rules football players (Dawson 2005) and effects of pre-exercise and post-exercise stretching by physically active people participating in self-selected recreational physical activity (Jamtvedt 2010).

Three studies examined the effect of stretching before exercise (High 1989; Johansson 1999; Wessel 1994a), seven studies investigated the effects of stretching after exercise (Dawson 2005; Gulick 1996; High 1989; Maxwell 1988; McGlynn 1979; Terry 1985; Wessel 1994b), one study investigated the effects of stretching before exercise in one intervention group and after exercise in another intervention group (Marques 2006), and one study investigated effects of stretching before and after exercise in the same intervention group (Jamtvedt 2010). The stretch was of the knee extensor muscles (Buroker 1989; High 1989; Maxwell 1988; Terry 1985), hamstring muscles (Johansson 1999; Wessel 1994a; Wessel 1994b), ankle plantarflexor muscles (Buroker 1989), wrist extensor muscles (Gulick 1996), elbow flexor muscles (Marques 2006; McGlynn 1979), or lower limb and trunk muscles (Dawson 2005; Jamtvedt 2010). All but one of the studies examined the effects of static stretching; one study examined the effects of a ("hold-relax") PNF technique (Terry 1985). Three studies involved repeated stretching sessions (Buroker 1989: 12 sessions; Jamtvedt 2010: before and after all physical activity for 12 weeks; McGlynn 1979: 5 sessions). The duration of stretching applied in a single session ranged from 40 seconds (Terry 1985) to 900 seconds (Jamtvedt 2010).

In eight studies (Buroker 1989; Dawson 2005; Gulick 1996; High 1989; Jamtvedt 2010; Marques 2006; Maxwell 1988; McGlynn

1979), participants were randomly allocated to stretch and control conditions. One of these (Dawson 2005) was a cross-over study in which each participant experienced both the stretch and control conditions. Four studies (Johansson 1999; Terry 1985; Wessel 1994a; Wessel 1994b) stratified by participant and randomised legs to stretch and control conditions. Thus five studies (the single cross-over study and the four studies that compared right and left legs) provided paired comparisons. In all but two of these insufficient data were provided in the report to take account of the paired nature of the data. Both of the studies by Wessel and Wan (Wessel 1994a; Wessel 1994b) provided confidence intervals for the differences between legs, making it possible to account for the paired nature of the data, but the estimates from an analysis which accounted for the paired nature of the data were less precise (Wessel 1994a) or had similar precision (Wessel 1994b) to estimates obtained by ignoring the paired nature of the data. Hence the data were treated as if they were independent. Consequently, data from all five studies with paired comparisons (Dawson 2005; Johansson 1999; Terry 1985; Wessel 1994a; Wessel 1994b) were treated as independent. Please see the [Characteristics of included studies](#) for further details.

Excluded studies

Fifteen studies were excluded (*see the Characteristics of excluded studies*). Most exclusions were because the study was not a randomised or quasi-randomised trial, or because the study intervention was not a pre-exercise or post-exercise stretching technique.

Risk of bias in included studies

Figure 1 summarises the risk of bias assessments. All included studies were described as randomised, but adequate methods of sequence generation were only reported in four trials. Allocation concealment was certain in Jamtvedt 2010 alone. There was no blinding in any studies: this is because it is difficult or impossible in studies of stretching to blind the person supervising the stretching exercises and the person doing the stretches and as the outcome (soreness) must be self-reported, the assessor cannot be blinded. Only one study clearly indicated that analysis was by intention-to-treat (Jamtvedt 2010). In one study (Dawson 2005), analysis was explicitly not by intention-to-treat. Only two studies explicitly reported completeness of follow-up: Dawson 2005 reported data were available for 31 of 34 player-conditions and Jamtvedt 2010 reported that participants provided a mean of 10.1 (SD 3.4) diary reports when they were asked to provide 12 reports. Two further studies (High 1989; Terry 1985) provided data that implied follow-up was complete. It is likely that follow-up was complete or nearly complete in most or all of the other studies because the follow-up period was very short (one week or less) and outcome measures were not onerous.

Figure 1. Risk of bias summary: review authors' judgements about each risk of bias item for each included study.

	Random sequence generation (selection bias)	Allocation concealment (selection bias)	Blinding (performance bias and detection bias)	Incomplete outcome data (attrition bias)	Selective reporting (reporting bias)	Other bias
Buroker 1989	?	?	—	?	?	?
Dawson 2005	?	—	—	?	?	—
Gulick 1996	?	?	—	+	?	?
High 1989	?	?	—	?	?	?
Jamtvedt 2010	+	+	—	—	+	?
Johansson 1999	?	?	—	?	?	?
Marques 2006	+	?	—	+	?	?
Maxwell 1988	?	?	—	?	?	?
McGlynn 1979	?	?	—	?	?	?
Terry 1985	?	?	—	?	?	?
Wessel 1994a	+	?	—	?	?	?
Wessel 1994b	+	?	—	?	?	?

Taken together, the quality of evidence of effects of pre-exercise or post-exercise stretching on muscle soreness after exercise is low to moderate.

Effects of interventions

See: [Summary of findings for the main comparison](#) Pre-exercise stretching to prevent soreness; [Summary of findings 2](#) Post-exercise stretching for people doing unaccustomed exercise; [Summary of findings 3](#) Either pre- or post-exercise stretching for people doing unaccustomed exercise; [Summary of findings 4](#) Both pre-exercise and post-exercise stretching for people doing unaccustomed exercise

One study (Gulick 1996) reported only pooled data from all groups and another two (Marques 2006; Maxwell 1988) did not provide any data that could be used to calculate precision. The author of one of these reports (Marques 2006) provided us with raw data so we were able to estimate the precision of that study. However, the standard deviations (SDs) were remarkably small, averaging just 3.8 on a 100-point scale. These SDs were much smaller than those for any other study and about one sixth of the pooled estimate of the SD from the remaining studies which reported SDs at the same time points. As a consequence, the effect of including data from this study in the analyses would have been to give very large weights (range 50% to 96%, mean 74%) to a small study which had just 9 or 10 observations in each group. Consequently we decided to exclude these data from this study in the primary analysis. A sensitivity analysis was conducted in which data from this study were included.

There was a remarkable degree of consistency (homogeneity) of results across studies (I^2 for all pooled estimates was 0%). The pooled estimate was that pre-exercise stretching reduced soreness one day after exercise by, on average, half a point on a 100-point scale (mean difference (MD) -0.52, 95% CI -11.30 to 10.26, where negative effects indicate that stretching decreased soreness; 3 studies; [Analysis 1.1](#)). Post-exercise stretching reduced soreness one day after exercise by, on average, one point on a 100-point scale (MD -1.04, 95% CI -6.88 to 4.79; 4 studies; [Analysis 2.1](#)). Similar effects were evident up to three days after exercise.

As there was no sign of statistical heterogeneity, estimates from pre-exercise and post-exercise studies were pooled. The pooled estimates indicate that pre-exercise and post-exercise stretching reduces soreness, on average, by one point on a 100-point scale at one day (MD -0.93, 95% CI -6.05 to 4.20; 7 studies; [Analysis 3.1](#)), increases soreness by one point on a 100-point scale at two days (MD 1.04, 95% CI -4.14 to 6.22; 7 studies; [Analysis 3.2](#)) and has no effect on soreness at three days (MD -0.28; 95% CI -6.79 to 6.22; 5 studies; [Analysis 3.3](#)).

The only large study (Jamtvedt 2010) examined effects of stretching before and after all episodes of physical activity undertaken over a 12 week period by people undertaking recreational exercise. [Jamtvedt 2010](#) found stretching reduced the intensity of the worst soreness experienced over a week by, on average, four points on a 100-point scale (MD -3.80; 95% CI -5.17 to -2.43; [Analysis 4.1](#)). [Jamtvedt 2010](#) reported that the risk of experiencing bothersome soreness in a week was 24.6% in the stretch group and 32.3% in the control group (OR 0.69; 95% CI 0.59 to 0.82; [Analysis 4.2](#)). In the only other study of the effects of stretching on soreness associated with recreational activity, [Dawson 2005](#) found that post-exercise stretching increased muscle soreness by, on average, three points on a 100-point scale (95% CI -7 to 13) at 15 hours and by four points (95% CI -5 to 13) at 48 hours after an Australian rules football match.

Similar estimates of effects were obtained in the only study that tested PNF stretching. [Terry 1985](#) showed the effect of PNF stretching was to reduce soreness by one point (95% CI -13 to 11) on a 100-point scale at both one and two days after exercise. A sensitivity analysis investigated the effects of including the data from one study with very small SDs (Marques 2006): see the first paragraph of this section for an explanation. Including data from that [Marques 2006](#) yielded slightly larger estimates of the effects of stretching: the pooled estimates of the effects of pre-exercise or post-exercise stretching were that stretching reduced soreness, on average, by six points on a 100-point scale (95% CI -9 to -4) one day after exercise and by two points (95% -6 to 1) two days after exercise.

ADDITIONAL SUMMARY OF FINDINGS *[Explanation]*

Post-exercise stretching for people doing unaccustomed exercise					
Patient or population: people doing unaccustomed exercise Settings: any Intervention: post-exercise stretching					
Outcomes	Illustrative comparative risks* (95% CI)		Relative effect (95% CI)	No of Participants (studies)	Quality of the evidence (GRADE)
	Assumed risk	Corresponding risk			
	Control	post-exercise stretching			
Soreness on day 1 Scale from: 0 to 100. Follow-up: 1 day	The mean soreness on day 1 in the control groups was 29	The mean soreness on day 1 in the intervention groups was 1.04 point lower (6.88 lower to 4.79 higher)		127 (4 studies)	⊕⊕○○ low ^{1,2,3}
Soreness on day 2 Scale from: 0 to 100. Follow-up: 2 days	The mean soreness on day 2 in the control groups was 35	The mean soreness on day 2 in the intervention groups was 1.12 points higher (4.63 lower to 6.87 higher)		158 (5 studies)	⊕⊕○○ low ^{1,2,3}
*The assumed risk for the 'control' group is based on the mean value of the results for all control groups in the trials included in each analysis. The corresponding risk (and its 95% confidence interval) is based on the assumed risk in the comparison group and the relative effect of the intervention (and its 95% CI). CI: Confidence interval;					

GRADE Working Group grades of evidence

High quality: Further research is very unlikely to change our confidence in the estimate of effect.

Moderate quality: Further research is likely to have an important impact on our confidence in the estimate of effect and may change the estimate.

Low quality: Further research is very likely to have an important impact on our confidence in the estimate of effect and is likely to change the estimate.

Very low quality: We are very uncertain about the estimate.

- ¹ Studies were randomised but little or no information was provided about allocation concealment, loss to follow-up or analysis by intention to treat.
- ² Most or all of these studies involved inducing muscle soreness using a standardised exercise protocol. Such studies provide an indirect evaluation of the effects of stretching on soreness caused by recreational exercise.
- ³ There were too few studies to reliably assess risk of publication bias or other sources of small study bias.

Either pre-exercise or post-exercise stretching for people doing unaccustomed exercise					
Patient or population: people doing unaccustomed exercise Settings: any Intervention: either pre- or post-exercise stretching					
Outcomes	Illustrative comparative risks* (95% CI)		Relative effect (95% CI)	No of Participants (studies)	Quality of the evidence (GRADE)
	Assumed risk	Corresponding risk			
	Control	either pre- or post-exercise stretching			
Soreness on day 1 Scale from: 0 to 100. Follow-up: 1 day	The mean soreness on day 1 in the control groups was 36	The mean soreness on day 1 in the intervention groups was 0.93 points lower (6.05 lower to 4.20 higher)		197 (7 studies)	⊕⊕○○ low ^{1,2,3}
Soreness on day 2 Scale from: 0 to 100. Follow-up: 2 days	The mean soreness on day 2 in the control groups was 38	The mean soreness on day 2 in the intervention groups was 1.04 higher (4.14 lower to 6.22 higher)		198 (7 studies)	⊕⊕○○ low ^{1,2,3}
*The assumed risk for the 'control' group is based on the mean value of the results for all control groups in the trials included in each analysis. The corresponding risk (and its 95% confidence interval) is based on the assumed risk in the comparison group and the relative effect of the intervention (and its 95% CI). CI: Confidence interval;					
GRADE Working Group grades of evidence High quality: Further research is very unlikely to change our confidence in the estimate of effect. Moderate quality: Further research is likely to have an important impact on our confidence in the estimate of effect and may change the estimate. Low quality: Further research is very likely to have an important impact on our confidence in the estimate of effect and is likely to change the estimate. Very low quality: We are very uncertain about the estimate.					

-
- ¹ Studies were randomised but little or no information was provided about allocation concealment, loss to follow-up or analysis by intention to treat.
 - ² Most or all of these studies involved inducing muscle soreness using a standardised exercise protocol. Such studies provide an indirect evaluation of the effects of stretching on soreness caused by recreational exercise.
 - ³ There were too few studies to reliably assess risk of publication bias or other sources of small study bias.

Both pre-exercise and post-exercise stretching for people doing unaccustomed exercise					
Patient or population: patients with people doing unaccustomed exercise Settings: any Intervention: both pre- and post-exercise stretching					
Outcomes	Illustrative comparative risks* (95% CI)		Relative effect (95% CI)	No of Participants (studies)	Quality of the evidence (GRADE)
	Assumed risk	Corresponding risk			
	Control	Both pre- and post-exercise stretching			
Soreness in preceding week Scale from: 0 to 100.	The mean soreness in preceding week in the control groups was 25	The mean soreness in preceding week in the intervention groups was 3.80 points lower (5.17 to 2.43 lower) ¹		2377 (1 study)	⊕⊕⊕○ moderate ^{2,3}
Risk of bothersome soreness in preceding week Self-report (yes/no)	Study population See comment	See comment	OR 0.69 (0.59 to 0.82) ¹	0 (1 study)	⊕⊕⊕○ moderate ²
	Medium risk population				
	323 per 1000	248 per 1000 (220 to 281)			
*The assumed risk for the 'control' group is based on the results for the control groups in the only trial included in this analysis. The corresponding risk (and its 95% confidence interval) is based on the assumed risk in the comparison group and the relative effect of the intervention (and its 95% CI). CI: Confidence interval; OR: Odds ratio;					

GRADE Working Group grades of evidence

High quality: Further research is very unlikely to change our confidence in the estimate of effect.

Moderate quality: Further research is likely to have an important impact on our confidence in the estimate of effect and may change the estimate.

Low quality: Further research is very likely to have an important impact on our confidence in the estimate of effect and is likely to change the estimate.

Very low quality: We are very uncertain about the estimate.

¹ The trial was a longitudinal study with repeated (non-independent) observations on individuals. The estimate provided in the trial report is used here.

² Significant loss to follow-up. However most subjects provided some follow-up data and longitudinal method of analysis deals optimally with missing data.

³ Only one study.

DISCUSSION

Summary of main results

When this review was first published it concluded the available evidence from randomised studies indicated that stretching soon before or soon after exercise does not produce important reductions in muscle soreness in the few days following exercise (Herbert 2007). That conclusion remains unchanged in this update of the review.

The primary analysis indicated that stretching has near-zero effects on soreness. Point estimates of the effect were that the average effect of stretching was to reduce soreness by up to four points on a 100-point scale. Inclusion of data from a study with very small standard deviations increased estimates slightly. The estimates were quite precise so, in the primary analyses, it was possible to rule out the possibility that the average effect of pre-exercise or post-exercise stretching was to reduce soreness by more than six points on a 100-point scale.

This update of the review incorporates two new studies, including one very large randomised trial with 2377 participants. The new trial provides an estimate of the effects of stretching both before and after exercise. The estimate is that stretching reduced soreness by, on average, four points on a 100-point scale. This effect, though statistically significant, is very small.

Overall completeness and applicability of evidence

All but two of the eligible studies were laboratory-based. These studies assessed whether supervised stretching reduced the soreness produced by standardised exercise involving eccentric contractions of a particular muscle group. Such studies can be considered to have an “explanatory” orientation (Herbert 2009) because they tell us about what the effects of stretching could be when administered under ideal conditions. It might have been expected that the effects of stretching would be smaller in community settings. In community settings, unlike the laboratory settings of these studies, stretching is not usually supervised so it may be carried out suboptimally. Also, the everyday experience of muscle soreness may be across many muscle groups, and it may be difficult to effectively stretch all of the muscle groups that might become sore. These considerations suggest the effects of stretching might be smaller in community settings than in laboratory settings. However the estimates from 10 laboratory-based studies and two community-based study suggest there are similar (near-zero) effects in both settings.

Quality of the evidence

These conclusions are based on studies that rate poorly or moderately on conventional scales of study quality. Only one of the reviewed studies explicitly concealed allocation, none were blinded,

few reported low rates of loss to follow-up and only one explicitly analysed by intention to treat. Nonetheless there are some reasons to believe that the findings may not be seriously biased: First, there is a high degree of consistency in estimates of the effect of stretching, both across studies and across comparisons. Also, biases due to non-concealment of allocation, lack of blinding, loss to follow-up and failure to analyse by intention-to-treat typically produce exaggerated effects (Chalmers 1983; Colditz 1989; Moher 1998; Schulz 1995) yet this review found consistent evidence of near-zero effects. The high degree of consistency of estimates across studies suggests that alternative weightings (based, for example, on different approaches to estimating precision from studies that used within-participant designs) would produce very similar pooled estimates of effects.

Other comments

The practice of stretching before or after exercise may have been arisen because of the mistaken belief, prevalent in the 1960s and 1970s, that muscle soreness was due to muscle spasm (de Vries 1961). This view has since been discredited (Bobbett 1986). Contemporary beliefs about the mechanisms of muscle soreness do not suggest any means by which stretching before or after exercise could reduce the soreness that follows that exercise session. However if, as has been proposed, muscle soreness is due to excessive elongation of some sarcomeres within muscle fibres (Proske 2001), it is conceivable that any intervention which increased the number of sarcomeres in series in muscle fibres, or which increased the length or compliance of tendons, could reduce sarcomere strains and lessen muscle damage associated with unaccustomed eccentric muscle contractions. This suggests the possibility that stretching may be more effective for people who have very short muscles, or if the stretching is repeated for weeks or months or years. The longest duration of stretching investigated in any of the trials conducted to date (Jamtvedt 2010) is 12 weeks. No randomised trials have specifically investigated interactions between effects of stretching and flexibility.

The best available evidence indicates that stretching does not reduce muscle soreness. However, there are other justifications for stretching. Some evidence suggests that once muscle soreness has developed, stretching may provide a transient relief of soreness (Reisman 2005). Some people stretch to reduce risk of injury, others stretch to enhance athletic or sporting performance, and yet others stretch because it gives them a sense of well-being. The current review does not provide any evidence of an effect or otherwise of stretching on risk of injury, performance or well-being.

AUTHORS' CONCLUSIONS

Implications for practice

The available evidence from randomised trials suggests that stretching before or after exercise does not produce important reductions in soreness in healthy adults.

Stretching could have other benefits. For example, it is widely believed that stretching can reduce injury risk, improve sporting performance and produce a feeling of readiness for exercise. People who are concerned about these outcomes should also consider the evidence of effects of stretching on these outcomes when deciding whether or not to stretch.

Implications for research

Arguably, the findings of this review are clear enough that further research into the effects of stretching on muscle soreness is not necessary. We see some merit in this view. There may be some merit in conducting further trials of the effects of longer-term stretching

programs and trials of stretching for people with reduced flexibility.

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* Indicates the major publication for the study

CHARACTERISTICS OF STUDIES

Characteristics of included studies [ordered by study ID]

Buroker 1989

Methods	Randomised (between participants) design (three groups).
Participants	Setting: laboratory, USA 23 participants (16 men and 7 women) aged 18-33 years with no known contraindications to exercise. None was highly physically trained and none were extremely active in the six weeks preceding the study
Interventions	Control group (N = 8): no stretching. One stretch group 1 (N = 7): stretched left knee extensors after exercise. A second stretch group (N = 8): stretched left knee extensors and right plantarflexors after exercise. Stretches (10 30-second static stretches, each separated by 10 seconds rest, to noticeable tension or slight discomfort) were performed at 2-hour intervals for the first 24 hours and 4-hour intervals for the following 48 hours except that participants were allowed to sleep for 8 hours each night
Outcomes	The following measures were taken on days 1, 2 and 3: 1. Soreness on a 0 to 6 scale (0 = no pain, light pain only on palpation, 4 = moderate pain, stiffness and/or weakness, especially on movement, 6 = severe pain limiting range of movement)
Notes	Data from the two stretch groups were pooled. Stretched after exercise. Soreness induced with 20 minutes of step-down exercise.

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Unclear risk	Subjects were randomly assigned to one of three groups..."
Allocation concealment (selection bias)	Unclear risk	No description of concealment procedure.
Blinding (performance bias and detection bias) All outcomes	High risk	Blinding of participants and assessors not possible.
Incomplete outcome data (attrition bias) All outcomes	Unclear risk	No data provided on flow of participants through trial.
Selective reporting (reporting bias)	Unclear risk	No protocol available.
Other bias	Unclear risk	Unclear if analysed by intention to treat.

Dawson 2005

Methods	Randomised (within participants, cross-over) design.
Participants	17 semi-professional Australian rules football players.
Interventions	Setting: field, Australia Stretch condition: 15 minutes of supervised gentle stretching of the legs and back. This involved "2-3 repeats of 30 s held stretches across several muscle groups and joints". Control condition: no recovery procedures.
Outcomes	The following measures were taken at 15 and 48 hours: 1. Soreness ("how do you feel today?") measured on a 1 to 7 scale anchored at "very, very good" and "very, very sore"
Notes	Part of a larger study in which each participant was exposed on 12 occasions, each separated by one week, to one of four interventions (control, stretch, pool walking, hot-cold regimen). Initial condition was allocated randomly and subsequent conditions by alternation. Stretch was before exercise. Soreness was induced by playing football. All participants underwent a "pool recovery program" on the two days following exercise

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Unclear risk	"the control, stretching and hot/cold recoveries were initially randomly allocated and then implemented in alternate fashion"
Allocation concealment (selection bias)	High risk	No description of concealment procedure. Allocation of the first condition for each participant was random but subsequent conditions were allocated in a fixed order, so only the first condition could have been concealed
Blinding (performance bias and detection bias) All outcomes	High risk	Blinding of participants and assessors not possible.
Incomplete outcome data (attrition bias) All outcomes	Unclear risk	No data provided on flow of participants through trial.
Selective reporting (reporting bias)	Unclear risk	No protocol available.
Other bias	High risk	Explicitly not analysed by intention-to-treat.

Gulick 1996

Methods	Randomised (between participants) design.
Participants	Setting: laboratory, USA ~22 untrained participants randomly drawn from a larger sample of 35 men and 38 women aged 21 to 40 years
Interventions	Stretch group (N ~ 11): static stretches of the wrist extensor muscles to “end range” for 10 minutes soon after exercise. Control group (N ~ 11): two placebo tablets after exercise and one further placebo tablet 24 hours later
Outcomes	The following measures were taken 24, 48 and 72 hours after exercise: 1. On a 10 cm visual analogue scale anchored at “no soreness at all” and “soreness as bad as it could be”
Notes	Part of a larger study in which participants were randomised to seven interventions (control, stretch, anti-inflammatory drug, intense exercise, ice massage, herbal ointment) . Stretch was after exercise. Soreness was induced by 15 sets of 15 isokinetic eccentric wrist extensor contractions

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Unclear risk	“We randomly assigned subjects to 1 of 7 groups”.
Allocation concealment (selection bias)	Unclear risk	No description of concealment procedure.
Blinding (performance bias and detection bias) All outcomes	High risk	Blinding of participants and assessors not possible.
Incomplete outcome data (attrition bias) All outcomes	Low risk	“Three subjects were lost through attrition”.
Selective reporting (reporting bias)	Unclear risk	No protocol available.
Other bias	Unclear risk	Unclear if analysed by intention to treat.

High 1989

Methods	Randomised (between participants) design.
Participants	Setting: laboratory, USA 30 untrained participants (stretch group, 7 men and 7 women, mean age 19.4 years; control group, 8 men and 8 women, mean age 20.4 years)

High 1989 (Continued)

Interventions	Stretch group: stretched the quadriceps muscles (5 stretches, twice each on each leg for 50 seconds) of both legs prior to exercise. Control group: did not stretch
Outcomes	The following measures were taken at 24 hour intervals for 5 days: 1. Soreness on a 0 to 6 scale (0 = "absence of soreness", 6 = "most severe pain possible from soreness")
Notes	Part of a larger study in which participants were randomised to four interventions (control, stretch, warm-up, warm-up and stretch). Stretch was before exercise. Soreness was induced by up to 80 minutes of step-down exercise

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Unclear risk	"Sixty two volunteers were randomly assigned to four groups."
Allocation concealment (selection bias)	Unclear risk	No description of concealment procedure.
Blinding (performance bias and detection bias) All outcomes	High risk	Blinding of participants and assessors not possible.
Incomplete outcome data (attrition bias) All outcomes	Unclear risk	No data provided on flow of participants through trial.
Selective reporting (reporting bias)	Unclear risk	No protocol available.
Other bias	Unclear risk	Unclear if analysed by intention to treat.

Jamtvedt 2010

Methods	Randomised (between participants) design. Study was internet-based
Participants	Setting: field, mainly Australia and Norway 2377 adults aged over 18 (mean age 40, ~63% female). Participants regularly engaged in physical activity at least once per week (median 4 times)
Interventions	Stretch group (n = 1220): stretched 7 muscle groups (gastrocnemius, hip adductors, hip flexors, hamstrings, rectus femoris, hip external rotators, truck rotators) before and after vigorous exercise. Each stretch was held for at least 30 seconds. Stretch intensity was such that participants felt a strong but not painful stretch Control group (n = 1157): did not stretch lower limb or trunk muscles

Outcomes	The following measures were taken weekly for the 12-week study period. 1. Whether or not the participant had experienced bothersome soreness in the legs, buttocks or back in the preceding week. 2. Rating of the worst soreness experienced in the legs, buttocks or back in the preceding week from 0 (no soreness at all) to 10 (worst imaginable soreness)	
Notes	The entire study (recruitment, randomisation, instructions on the intervention, measurement of outcomes) was conducted over the internet	
<i>Risk of bias</i>		
Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	“The randomisation schedule was unrestricted (no stratification or blocking) and was administered by computer.”
Allocation concealment (selection bias)	Low risk	“This ensured the randomisation sequence was concealed”.
Blinding (performance bias and detection bias) All outcomes	High risk	Blinding of participants and assessors not possible.
Incomplete outcome data (attrition bias) All outcomes	High risk	Outcomes were reported weekly for 12 weeks. While most participants (98%%) provided at least one weekly report of soreness, outcome reporting was incomplete (72% of all required reports)
Selective reporting (reporting bias)	Low risk	Protocol was published on the internet (http://www.kunnskapssenteret.no/The%20Stretching%20Trial) prior to commencement of study. Primary and secondary outcomes in the protocol correspond to those in the trial report
Other bias	Unclear risk	Unclear if analysed by intention to treat.

Johansson 1999

Methods	Randomised (between legs) design.
Participants	Setting: laboratory, Sweden 10 health female physiotherapy students (mean 24 (SD 3) years, 62 (SD 3) kg, 169 (SD 5) cm), not taking anti-inflammatory medication, with no lower limb musculoskeletal symptoms, had not participated in weight training for more than 3 hours per week, and were not experiencing DOMS at the time of testing

Johansson 1999 (Continued)

Interventions	Stretch leg: four 20-second “hamstring hurdle” stretches, with 20 second rests between stretches. Stretch intensity was sufficient to induce a strong feeling of stretch but not pain. Control leg: the hamstring muscles on the other side of the body were not stretched	
Outcomes	The following measures were taken 24, 48 and 96 hours after exercise: 1. Hamstring muscle soreness on a 100 mm visual analogue scale (0 = “no soreness”, 100 = “worst possible soreness”)	
Notes	Stretch was before exercise. Soreness was induced with 10 sets of 10 isokinetic knee flexor contractions	
<i>Risk of bias</i>		
Bias	Authors’ judgement	Support for judgement
Random sequence generation (selection bias)	Unclear risk	“Before exercise subjects performed an autostretching procedure for the right or left hamstring in random order”
Allocation concealment (selection bias)	Unclear risk	No description of concealment procedure.
Blinding (performance bias and detection bias) All outcomes	High risk	Blinding of participants and assessors not possible.
Incomplete outcome data (attrition bias) All outcomes	Unclear risk	No data provided on flow of participants through trial.
Selective reporting (reporting bias)	Unclear risk	No protocol available.
Other bias	Unclear risk	Unclear if analysed by intention to treat.

Marques 2006

Methods	Randomised (between participants) design.	
Participants	Setting: laboratory, Brazil 30 healthy adult male army recruits, mean age 18 years.	
Interventions	Group 1: stretched elbow flexor muscles bilaterally before eccentric and concentric strength exercise. Group 2: stretched elbow flexor muscles bilaterally after eccentric and concentric strength exercise. Group 3: did not stretch Stretches were three 30-second stretches conducted according to a protocol developed by	

Marques 2006 (Continued)

	Weldon & Hill (Weldon SM, Hill RH (2003). The efficacy of stretching for prevention of exercise-related injury: a systematic review of the literature. Manual Therapy 8: 141-150)	
Outcomes	Soreness was measured 24, 48 and 72 hours after the exercise on a 10 cm visual analogue scale	
Notes	No SD data provided in trial report. However the author provided raw data on request	
<i>Risk of bias</i>		
Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	"Participants were randomly allocated through ballot to three different groups"
Allocation concealment (selection bias)	Unclear risk	No description of concealment procedure.
Blinding (performance bias and detection bias) All outcomes	High risk	Blinding of participants and assessors not possible.
Incomplete outcome data (attrition bias) All outcomes	Low risk	No data provided on flow of participants through trial in trial report, but author provided data on request. Data were obtained from all subjects at 24 hours, but one participant of each of the two intervention groups (10% of each group) was lost to 48 and 72 hour follow-ups
Selective reporting (reporting bias)	Unclear risk	No protocol available.
Other bias	Unclear risk	Unclear if analysed by intention-to-treat.

Maxwell 1988

Methods	Randomised (between participants) design.
Participants	Setting: laboratory, Australia ~14 female first-year college students randomly drawn from a larger sample of 20 participants aged 18-25 years (mean 19.4) and with varying histories of physical activity
Interventions	Stretch group (N ~ 6): participants lay in prone with the hips in a neutral position. A static stretch was applied to the quadriceps to the point of discomfort and held for 15 seconds. Control group (N ~ 6): participants did not stretch.

Maxwell 1988 (Continued)

Outcomes	The following measures were taken 24, 48 and 72 hours after exercise: 1. Pain experienced in the quadriceps muscle whilst descending a flight of stairs rated on a 10 cm visual analogue scale anchored at 0 ("no pain") and 10 ("severe pain")
Notes	Stretch was after exercise. Soreness was induced with 15 minutes of step-down exercise.

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Unclear risk	"After the exercise, subjects were randomly allocated to one of three treatment groups"
Allocation concealment (selection bias)	Unclear risk	No description of concealment procedure.
Blinding (performance bias and detection bias) All outcomes	High risk	Blinding of participants and assessors not possible.
Incomplete outcome data (attrition bias) All outcomes	Unclear risk	No data provided on flow of participants through trial.
Selective reporting (reporting bias)	Unclear risk	No protocol available.
Other bias	Unclear risk	Unclear if analysed by intention-to-treat.

McGlynn 1979

Methods	Randomised (between participants) design
Participants	Setting: laboratory, USA 24 men aged 18 to 26 years who had not engaged in any systematic activity with their non-dominant arms in the past 30 days
Interventions	Stretch group (N = 12): participants "hyperextended" the "forearm" for two minutes before releasing the stretch for one minute. This sequence was repeated four times each at 6, 25, 30, 49 and 54 hours after exercise. Control group (N = 12): participants did not stretch.
Outcomes	The following measures were taken 24, 48 and 72 hours after exercise: 1. Pain rated on a 30-point scale, where 0 to 10 was "mild" pain, 11 to 20 was "moderate" pain, and 21 to 30 was "severe" pain
Notes	Part of a larger study in which participants were randomised to three interventions (control, stretch, biofeedback). Stretch was after exercise.

McGlynn 1979 (Continued)

	Soreness was induced with eccentric elbow flexor exercises performed to failure	
<i>Risk of bias</i>		
Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Unclear risk	“The subjects were assigned at random to one of three groups”
Allocation concealment (selection bias)	Unclear risk	No description of concealment procedure.
Blinding (performance bias and detection bias) All outcomes	High risk	Blinding of participants and assessors not possible.
Incomplete outcome data (attrition bias) All outcomes	Unclear risk	No data provided on flow of participants through trial.
Selective reporting (reporting bias)	Unclear risk	No protocol available.
Other bias	Unclear risk	Unclear if analysed by intention to treat.

Terry 1985

Methods	Randomised (between legs) design.	
Participants	Setting: laboratory, Australia 30 healthy participants, 15 men and 15 women, aged between 18 and 35 years. The modal level of exercise was three times per week	
Interventions	Stretch (hold-relax) leg: participants lay prone with a wedge under the thigh and the knee was actively flexed as far as possible. Then the participant performed an isometric contraction against manual resistance for 4 to 6 seconds. This was followed by two seconds of relaxation and then an 8 second passive stretch. This was repeated 5 times. Control leg: the other leg was not stretched.	
Outcomes	The following measures were taken on days 2 and 3: 1. “Pain” (or, variably, “pain, soreness or stiffness” or “soreness”) experienced that day during functional movements rated on a scale of 0 (“no pain”) to 7 (“unbearably painful”)	
Notes	Stretching was after exercise. Soreness was induced with 20 minutes of step-up and step-down exercise	
<i>Risk of bias</i>		
Bias	Authors’ judgement	Support for judgement

Terry 1985 (Continued)

Random sequence generation (selection bias)	Unclear risk	“At the conclusion of the step-test and after a standard warm-down and independent tester applied a hold-relax stretch to the randomly chosen experimental leg.”
Allocation concealment (selection bias)	Unclear risk	No description of concealment procedure.
Blinding (performance bias and detection bias) All outcomes	High risk	Blinding of participants and assessors not possible.
Incomplete outcome data (attrition bias) All outcomes	Unclear risk	No data provided on flow of participants through trial.
Selective reporting (reporting bias)	Unclear risk	No protocol available.
Other bias	Unclear risk	Unclear if analysed by intention-to-treat.

Wessel 1994a

Methods	Randomised (between legs) design	
Participants	Setting: laboratory, Canada 10 sedentary adults (5 women and 5 men) with no history of back or lower limb injury or disease, mean age 24.2 (SD 5.37), mean height 175.5 cm (SD 7.6) and mean mass 66.3 kg (8.9)	
Interventions	Stretch leg: 10 60-second stretches of the hamstring muscles prior to exercise. Control leg: not stretched.	
Outcomes	The following measures were taken at 12 hour intervals to 72 hours after exercise: 1. Pain on a 10 cm visual analogue scale while walking on an “even” surface	
Notes	Randomisation was by toss of a coin. Stretching was before exercise. Soreness was induced by 3 sets of 20 isokinetic eccentric knee flexor contractions	
<i>Risk of bias</i>		
Bias	Authors’ judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	“The side of control and experimental legs was randomly assigned by the flip of a coin, as was the order that they were to be exercised”
Allocation concealment (selection bias)	Unclear risk	No description of concealment procedure.

Wessel 1994a (Continued)

Blinding (performance bias and detection bias) All outcomes	High risk	Blinding of participants and assessors not possible.
Incomplete outcome data (attrition bias) All outcomes	Unclear risk	No data provided on flow of participants through trial.
Selective reporting (reporting bias)	Unclear risk	No protocol available.
Other bias	Unclear risk	Unclear if analysed by intention-to-treat.

Wessel 1994b

Methods	Randomised (between legs) design.
Participants	Setting: laboratory, Canada 10 sedentary adults (2 women and 8 men) with no history of back or lower limb injury or disease, mean age 25.2 (SD 3.36), mean height 171.2 cm (SD 7.29) and mean mass 61.3 kg (7.12)
Interventions	Stretch leg: 10 60-second stretches of the hamstring muscles after exercise. Control leg: not stretched.
Outcomes	The following measures were taken at 12 hour intervals to 72 hours after exercise: 1. Pain on a 10 cm visual analogue scale while walking on an "even" surface
Notes	Randomisation was by toss of a coin. Stretching was after exercise. Soreness was induced by 3 sets of 20 isokinetic eccentric knee flexor contractions

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	"The side of control and experimental legs was randomly assigned by the flip of a coin, as was the order that they were to be exercised"
Allocation concealment (selection bias)	Unclear risk	No description of concealment procedure.
Blinding (performance bias and detection bias) All outcomes	High risk	Blinding of participants and assessors not possible.
Incomplete outcome data (attrition bias) All outcomes	Unclear risk	No data provided on flow of participants through trial.

Wessel 1994b (Continued)

Selective reporting (reporting bias)	Unclear risk	No protocol available.
Other bias	Unclear risk	Unclear if analysed by intention-to-treat.

DOMS: delayed onset muscle soreness

s: second

SD: standard deviation

~: approximately

The numbers of participants given in the Participants and Interventions columns are the number of participants who received the stretch or control intervention. Thus, in studies with three or more arms the number of participants given is less than the total number of participants in the study. In Gulick 1996, 73 participants were randomised to 7 groups but the size of each group is not given, so sample sizes are given as “~11”. In Maxwell 1988, 20 participants were randomised to 3 groups but the size of each group is not given, so sample sizes are given as “~6”. In McGlynn 1979, 36 participants were randomised to 3 groups but the size of each group is not given, so sample sizes are given as “~12”.

Characteristics of excluded studies [ordered by study ID]

Study	Reason for exclusion
Abraham 1977	Stretching occurred 24 hours after exercise; not clear if conditions (stretch/flex) were randomised
Bale 1991	No stretch condition.
de Vries 1966	Not randomised (compared before and after stretch).
Jayaraman 2004	Stretching did not begin until 36 hours after exercise.
LaRoche 2006	Stretching not “soon before or soon after exercise”.
Lightfoot 1997	Plantarflexors were sore, but quadriceps and hamstring muscles were stretched (fails criterion of “stretching technique designed to prevent delayed-onset muscle soreness”)
Lund 1998	Not randomised (used historical controls).
Prentice 1982	Stretch combined with heat or cold; interventions 24 hours after exercise; did not measure soreness
Rahnama 2005	Stretching combined with jogging and walking and submaximal concentric contractions
Reilly 2002	Not clearly randomised; stretch combined with jogging and “shake-down”
Rhea 2009	Compares effects of static stretching with and without vibration
Robey 2009	Not randomised (confirmed in correspondence with the first author on 25 March 2010)

(Continued)

Rodenburg 1994	Intervention was a combination of warm-up, stretch and massage
Smith 1993	Compares effects of static and ballistic stretching.
Wang 2006	Did not measure pain or tenderness.

Characteristics of studies awaiting assessment *[ordered by study ID]*

[Buckenmeyer 1998](#)

Methods	Not known
Participants	Not known
Interventions	Not known
Outcomes	Not known
Status	Completed
Notes	Full article not located

[Costa 1997](#)

Methods	Not known
Participants	Not known
Interventions	Not known
Outcomes	Not known
Status	Completed
Notes	Full article not location

[Lin 1999](#)

Methods	Not known
Participants	Not known
Interventions	Not known
Outcomes	Not known

Lin 1999 (Continued)

Status	Completed
Notes	Full article not located

Lu 1992

Methods	Not known
Participants	Not known
Interventions	Not known
Outcomes	Not known
Status	Completed
Notes	Full articles not located

DATA AND ANALYSES

Comparison 1. Pre-exercise stretching

Outcome or subgroup title	No. of studies	No. of participants	Statistical method	Effect size
1 Soreness on day 1 (assessed 18-30 hours post-exercise)	3	70	Mean Difference (IV, Fixed, 95% CI)	-0.52 [-11.30, 10.26]
2 Soreness on day 2 (assessed 42-54 hours post-exercise)	2	40	Mean Difference (IV, Fixed, 95% CI)	0.72 [-11.20, 12.64]
3 Soreness on day 3 (assessed 66-78 hours post-exercise)	2	40	Mean Difference (IV, Fixed, 95% CI)	-2.50 [-15.82, 10.82]

Comparison 2. Post-exercise stretching

Outcome or subgroup title	No. of studies	No. of participants	Statistical method	Effect size
1 Soreness on day 1 (assessed 18-30 hours post-exercise)	4	127	Mean Difference (IV, Fixed, 95% CI)	-1.04 [-6.88, 4.79]
2 Soreness on day 2 (assessed 42-54 hours post-exercise)	5	158	Mean Difference (IV, Fixed, 95% CI)	1.12 [-4.63, 6.87]
3 Soreness on day 3 (assessed 66-78 hours post-exercise)	3	67	Mean Difference (IV, Fixed, 95% CI)	-0.03 [-7.49, 7.43]

Comparison 3. Either pre-exercise or post-exercise stretching

Outcome or subgroup title	No. of studies	No. of participants	Statistical method	Effect size
1 Soreness on day 1 (assessed 18-30 hours post-exercise)	7	197	Mean Difference (IV, Fixed, 95% CI)	-0.93 [-6.05, 4.20]
2 Soreness on day 2 (assessed 42-54 hours post-exercise)	7	198	Mean Difference (IV, Fixed, 95% CI)	1.04 [-4.14, 6.22]
3 Soreness on day 3 (assessed 66-78 hours post-exercise)	5	107	Mean Difference (IV, Fixed, 95% CI)	-0.28 [-6.79, 6.22]

Comparison 4. Both pre-exercise and post-exercise stretching

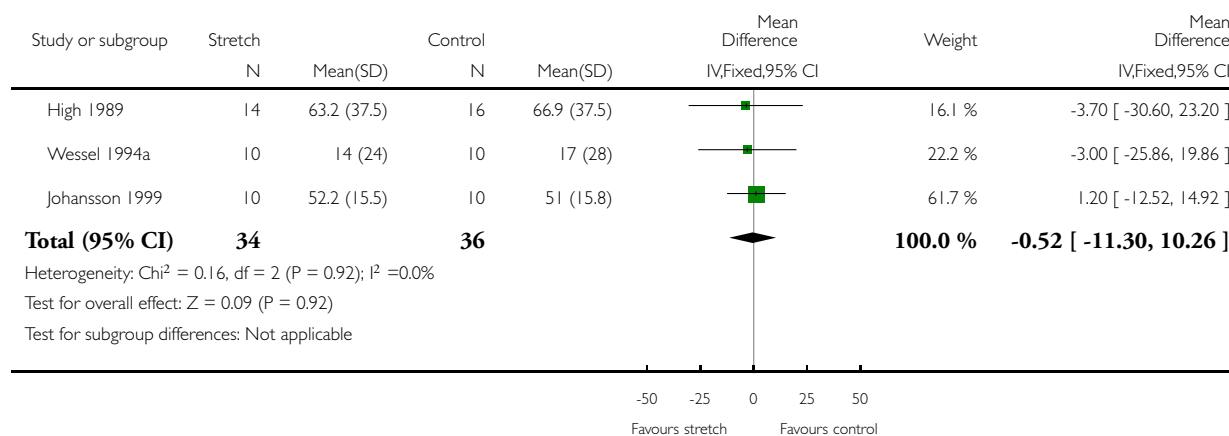
Outcome or subgroup title	No. of studies	No. of participants	Statistical method	Effect size
1 Soreness in preceding week	1	2323	Mean Difference (Fixed, 95% CI)	-3.8 [-5.17, -2.43]
2 Bothersome soreness	1	2323	Odds Ratio (Fixed, 95% CI)	0.69 [0.59, 0.82]

Analysis 1.1. Comparison 1 Pre-exercise stretching, Outcome 1 Soreness on day 1 (assessed 18-30 hours post-exercise).

Review: Stretching to prevent or reduce muscle soreness after exercise

Comparison: 1 Pre-exercise stretching

Outcome: 1 Soreness on day 1 (assessed 18-30 hours post-exercise)

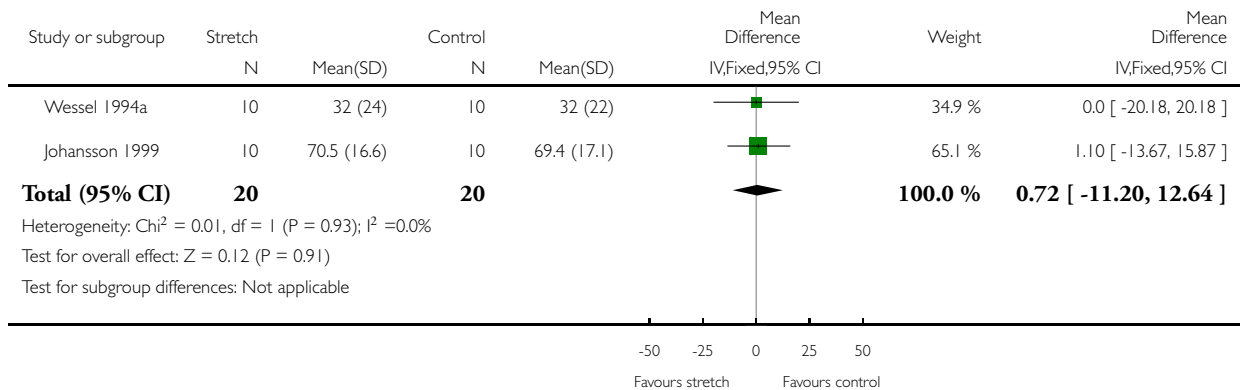


Analysis 1.2. Comparison 1 Pre-exercise stretching, Outcome 2 Soreness on day 2 (assessed 42-54 hours post-exercise).

Review: Stretching to prevent or reduce muscle soreness after exercise

Comparison: 1 Pre-exercise stretching

Outcome: 2 Soreness on day 2 (assessed 42-54 hours post-exercise)

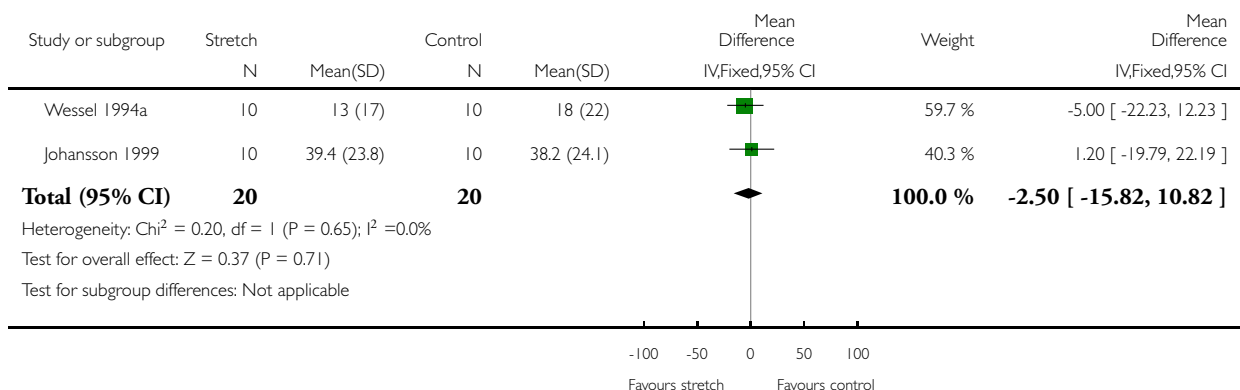


Analysis 1.3. Comparison 1 Pre-exercise stretching, Outcome 3 Soreness on day 3 (assessed 66-78 hours post-exercise).

Review: Stretching to prevent or reduce muscle soreness after exercise

Comparison: 1 Pre-exercise stretching

Outcome: 3 Soreness on day 3 (assessed 66-78 hours post-exercise)

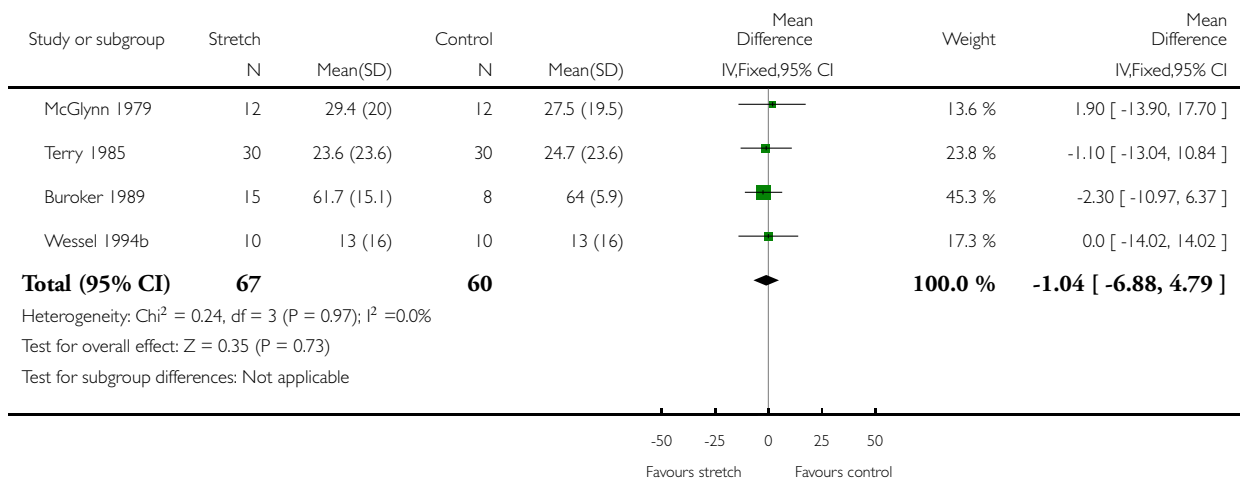


Analysis 2.1. Comparison 2 Post-exercise stretching, Outcome 1 Soreness on day 1 (assessed 18-30 hours post-exercise).

Review: Stretching to prevent or reduce muscle soreness after exercise

Comparison: 2 Post-exercise stretching

Outcome: 1 Soreness on day 1 (assessed 18-30 hours post-exercise)

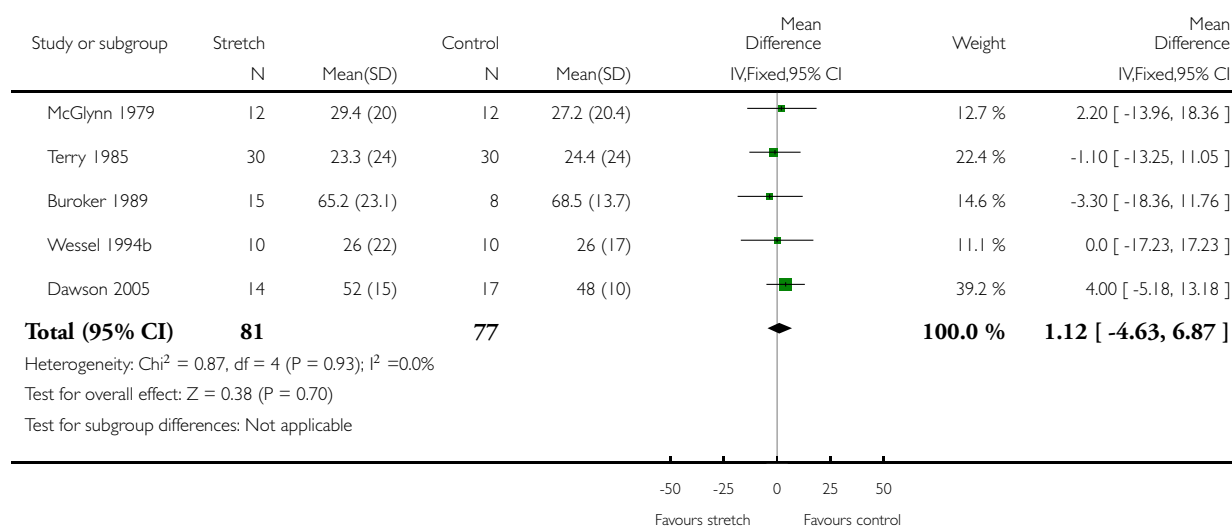


Analysis 2.2. Comparison 2 Post-exercise stretching, Outcome 2 Soreness on day 2 (assessed 42-54 hours post-exercise).

Review: Stretching to prevent or reduce muscle soreness after exercise

Comparison: 2 Post-exercise stretching

Outcome: 2 Soreness on day 2 (assessed 42-54 hours post-exercise)

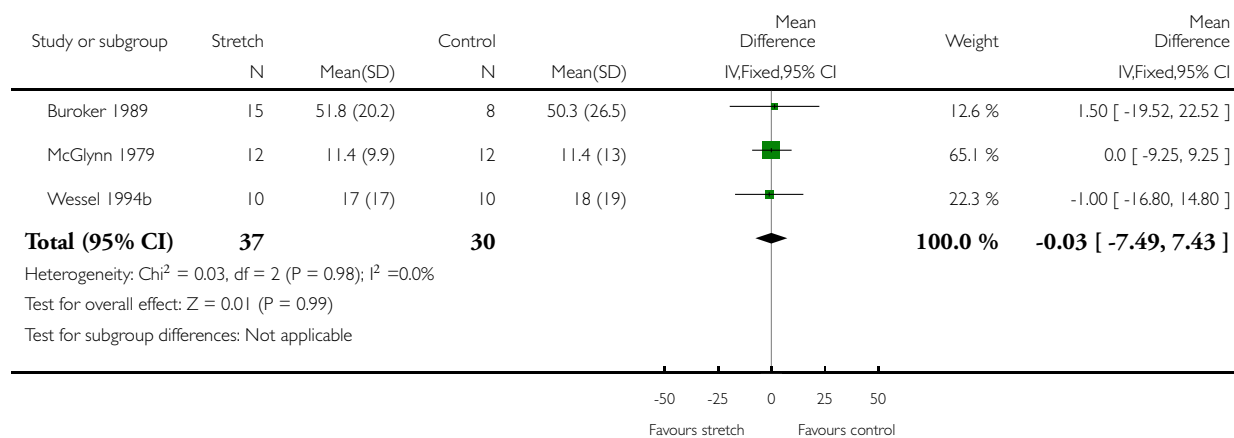


Analysis 2.3. Comparison 2 Post-exercise stretching, Outcome 3 Soreness on day 3 (assessed 66-78 hours post-exercise).

Review: Stretching to prevent or reduce muscle soreness after exercise

Comparison: 2 Post-exercise stretching

Outcome: 3 Soreness on day 3 (assessed 66-78 hours post-exercise)

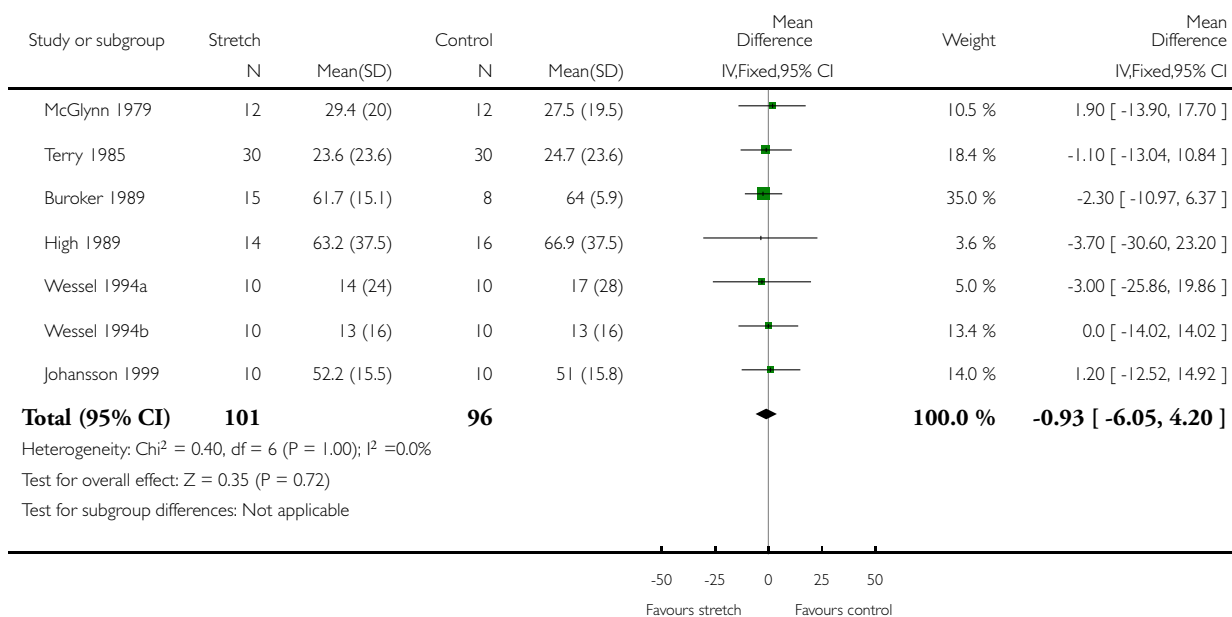


Analysis 3.1. Comparison 3 Either pre-exercise or post-exercise stretching, Outcome 1 Soreness on day 1 (assessed 18-30 hours post-exercise).

Review: Stretching to prevent or reduce muscle soreness after exercise

Comparison: 3 Either pre-exercise or post-exercise stretching

Outcome: 1 Soreness on day 1 (assessed 18-30 hours post-exercise)

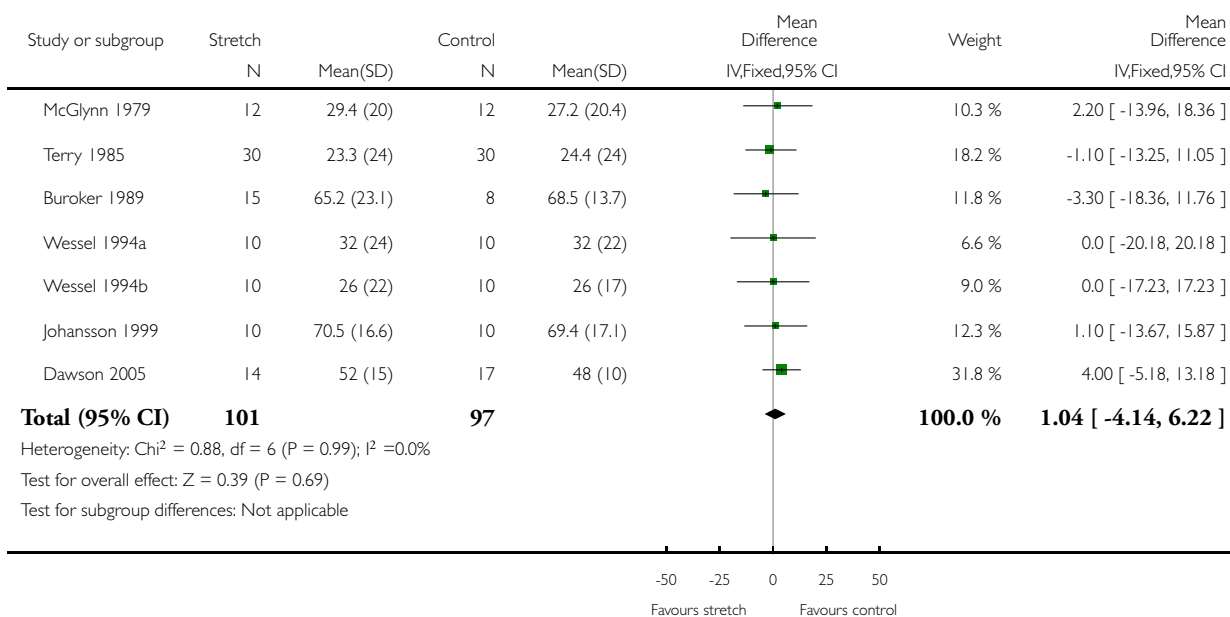


Analysis 3.2. Comparison 3 Either pre-exercise or post-exercise stretching, Outcome 2 Soreness on day 2 (assessed 42-54 hours post-exercise).

Review: Stretching to prevent or reduce muscle soreness after exercise

Comparison: 3 Either pre-exercise or post-exercise stretching

Outcome: 2 Soreness on day 2 (assessed 42-54 hours post-exercise)

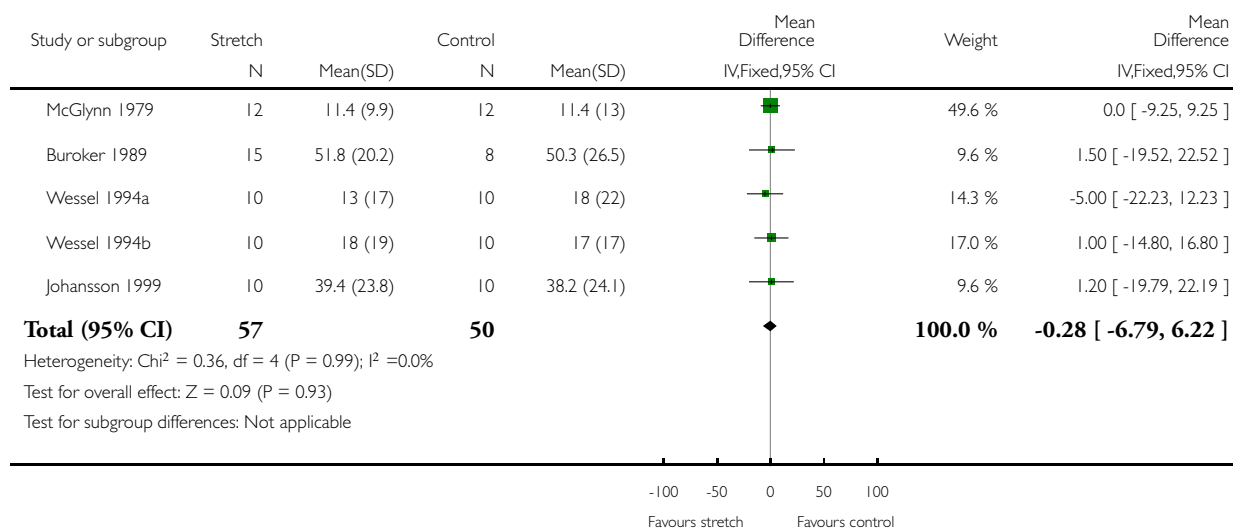


Analysis 3.3. Comparison 3 Either pre-exercise or post-exercise stretching, Outcome 3 Soreness on day 3 (assessed 66-78 hours post-exercise).

Review: Stretching to prevent or reduce muscle soreness after exercise

Comparison: 3 Either pre-exercise or post-exercise stretching

Outcome: 3 Soreness on day 3 (assessed 66-78 hours post-exercise)

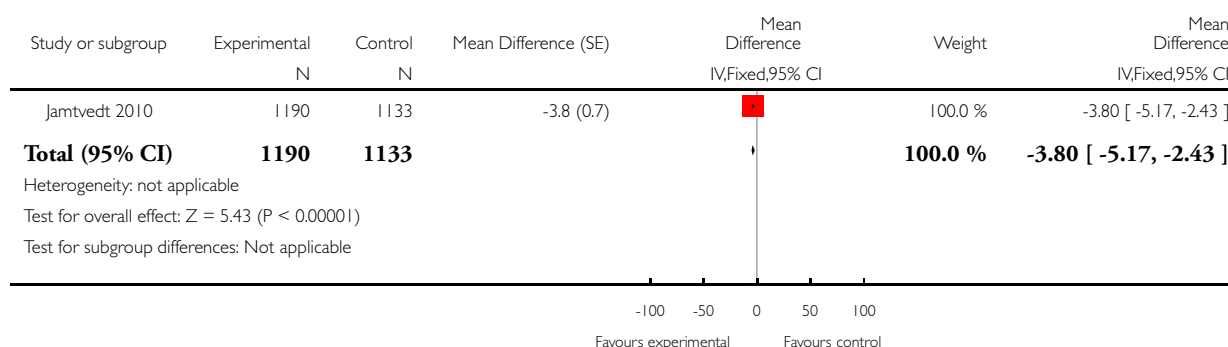


Analysis 4.1. Comparison 4 Both pre-exercise and post-exercise stretching, Outcome 1 Soreness in preceding week.

Review: Stretching to prevent or reduce muscle soreness after exercise

Comparison: 4 Both pre-exercise and post-exercise stretching

Outcome: 1 Soreness in preceding week

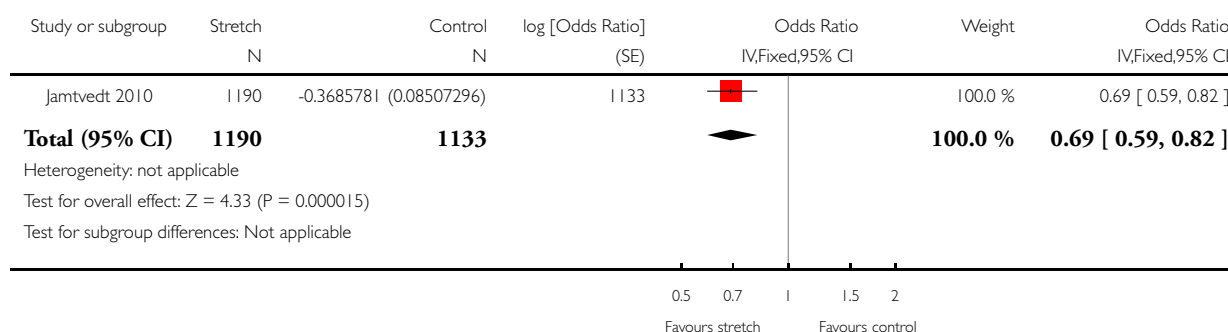


Analysis 4.2. Comparison 4 Both pre-exercise and post-exercise stretching, Outcome 2 Botherome soreness.

Review: Stretching to prevent or reduce muscle soreness after exercise

Comparison: 4 Both pre-exercise and post-exercise stretching

Outcome: 2 Botherome soreness



APPENDICES

Appendix I. Search strategies

Cochrane Central Register of Controlled Trials and MEDLINE (OVID WEB)

1. stretch\$.tw.
2. flexib\$.tw.
3. Pliability/
4. (range adj3 motion).tw.
5. "Range of Motion, Articular"/
6. (warmup or warm-up or warm up or cooldown or cool-down or cool down).tw.
7. or/1-6
8. (sore\$ adj3 musc\$).tw.
9. DOMS.tw.
10. exp Muscle Contraction/
11. Muscle, Skeletal/
12. or/10-11
13. Pain/
14. and/12-13
15. or/8-9,14
16. and/7,15

EMBASE (Embase.com)

1. stretch*
2. flexib*
3. 'pliability'/exp
4. 'range' AND 'motion'/exp
5. warmup OR 'warm up' OR warm AND up OR cooldown OR 'cool down' OR cool AND down
6. #1 OR #2 OR #3 OR #4 OR #5
7. sore* AND musc*
8. doms
9. 'muscle'/syn AND contraction
10. 'muscle'/syn AND skeletal
11. #9 OR #10
12. 'pain'/syn
13. #11 AND #12
14. #7 OR #8 OR #13
15. #6 AND #14
16. #15 AND [embase]/lim

CINAHL (via EBSCO)

1. stretch*
2. flexib*
3. pliability+
4. range N1 motion
5. warmup or warm-up or warm up or cooldown or cool-down or cool down
6. MH range of motion
7. S1 or S2 or S3 or S4 or S5 or S6
8. sore* N0 musc*
9. doms
10. muscle contraction+

11. MH muscle, skeletal
12. S10 or S11
13. MH pain
14. S12 and S13
15. S8 or S9 or S14
16. S7 and S15

SPORTDiscus (via EBSCO)

1. stretch*
2. flexib*
3. range N1 motion
4. (warmup or warm-up or warm up or cooldown or cool-down or cool down)
5. DE "JOINTS -- Range of motion" OR DE "JOINTS -- Hypermobility"
6. S1 or S2 or S3 or S4 or S5
7. sore* N0 musc*
8. doms
9. DE "MUSCLE contraction" OR DE "MUSCLE cramps" OR DE "MUSCLE tone"
10. DE "MUSCULOSKELETAL system" OR DE "BONE density" OR DE "BONES" OR DE "CONNECTIVE tissues" OR DE "JOINTS" OR DE "MUSCLES"
11. S9 or S10
12. DE "PAIN" OR DE "MUSCLE soreness"
13. S11 and S12
14. S7 or S8 or S13
15. S6 and S14

PEDro

The first search combined the following terms with "OR":

[in the Title/abstract field] stretch*, sore*, DOMS

The second search combined the following terms with "AND":

[in the Therapy field] stretching, mobilisation, manipulation, massage

[in the Subdiscipline field] sports

[in the Problem field] pain

WHAT'S NEW

Date	Event	Description
9 June 2011	New search has been performed	For this update, published in Issue 7, 2011, the following changes were made: 1. The search was updated to February 2010 2. Two new studies (Jamtvedt 2010; Marques 2006) were included. 3. In line with Collaboration policy, included trials were assessed using the 'Risk of bias' tool
9 June 2011	New citation required but conclusions have not changed	A new co-author was added to the review team.

HISTORY

Date	Event	Description
4 July 2008	Amended	Converted to new review format.

CONTRIBUTIONS OF AUTHORS

The protocol for the current review was written by Rob Herbert. Data collection, data extraction, quality assessment, data analysis and interpretation were carried out by Marcos de Noronha, Steve Kamper and Rob Herbert. Rob Herbert wrote the first draft of the review. Marcos de Noronha and Steve Kamper commented on drafts and approved the final version. Rob Herbert is guarantor for the review.

DECLARATIONS OF INTEREST

None known. However, as Rob Herbert is the contact investigator for [Jamtvedt 2010](#), data collection and risk of bias assessment of this trial was done independently by the two other authors.

SOURCES OF SUPPORT

Internal sources

- University of Sydney, Australia.

External sources

- No sources of support supplied

DIFFERENCES BETWEEN PROTOCOL AND REVIEW

The original protocol specified that study quality would be assessed using the Cochrane Bone, Joint and Muscle Trauma Group quality assessment tool. In keeping with the change in Cochrane Collaboration policy, the update of the review used instead the Cochrane 'Risk of bias' tool. In addition, GRADE was used to assess the quality of evidence.

In the first issue of the review, outcome data were extracted for outcomes at 12 hour intervals, rather than at 24 hour intervals. Extracting data at 12 hour intervals yielded little more data than when extracting data only at 24 hour intervals, substantially increased the number of tables and analyses, and had no effect on the findings of the review. Consequently, in the second round of review, only data from 24 hour intervals were included.

The protocol specified that a criterion for inclusion was that studies measured as outcomes either muscle soreness (pain) or tenderness (pain on palpation of the muscle). The first published version of the review did not identify any eligible studies which included only tenderness (not soreness) as an outcome. However, the second version identified one study that might have been considered to have measured tenderness ([Wang 2006](#)). We were uncertain whether this study could be said to have measured tenderness and, had it been included, it would have been the only study included in the review that provided only tenderness data. Moreover, we were uncertain about how data on tenderness would contribute to the objective of the review ("to determine the effect of stretching before or after exercise on the development of post-exercise muscle soreness"). Consequently, we decided to exclude studies which measured tenderness but not soreness.

INDEX TERMS

Medical Subject Headings (MeSH)

*Exercise; Muscle Stretching Exercises; Muscular Diseases [*prevention & control]; Pain [*prevention & control]; Randomized Controlled Trials as Topic

MeSH check words

Humans